Day 21 — Router & Multicast

Part A — Router

1. What Is a Router?

> **Router:** to **route the data to one branch based on a condition.** If you have multiple conditions, you can use a Router.

Example Scenario

Postman → Body → raw → XML containing `<status>active</status>`:

- if the status is **active** → send it in **one direction** (route);
- if the status is **not active** → send it to **another route**.

In this scenario we use a **Router**, **based on a condition**. *(Your trainer built this demo live in the system.)*

Message (status)	→ **Router**	Route 1: status = active	→ Receiver A
		Route 2: other condition	→ Receiver B
		Default route	→ End / Error path

2. Router Configuration — Processing Tab

Setting	Options / Examples
Expression Type	**XML** or **Non-XML**
Condition	XML → an **XPath** (e.g., `//status = 'active'`) · Non-XML → `${ ... }` expression (e.g., `${header.sender} = '1'`)
Default Route	The route taken when no other condition is true — **mandatory** (section 3)

XML vs. Non-XML Conditions

This condition **not only works with XML, but also works with non-XML conditions** — like **property (or) headers**.

Operators Available in Non-XML Conditions

Type	Operators
Comparison	`=` `!=` `>` `>=` `<` `<=`
Logical	`and` `or`
Text / list	`contains` `not contains` `in` `not in`
Pattern	`regex` `not regex`

All of the above operations work for the Router — e.g., `${header.sender} = '1'`.

Error: "Invalid format of condition expression value"

What does it mean? If we provide a **wrong format in a non-XML condition**, we get this error. In your trainer's example the cause is the **value**, which must be given in **single quotes** — e.g., `${header.RouterId} = '1'`.

> **Other common causes (verified in SAP Community threads):**
>
> - **Wrong capitalisation** — it must be lowercase `${property.X}` / `${header.X}`, not `${Property.X}`.
> - **Spaces inside a quoted value** — e.g., `contains 'my token string'` was reported to fail; workaround: compare against a property, or avoid spaces in the literal.
> - **Stale error** — after fixing the expression, **save the iFlow**; the error message clears only on save.

3. The Default Route — "Very Important from an Interview Perspective"

> **If you are using a Router, you must define the Default Route.**

Why? CPI is a middleware tool — it works as a **step-by-step process flow**. **If a process starts, it must end somewhere** — that is a basic functionality of CPI.

Question: if you receive a status **other than "active"** (say "inactive", or something unexpected), **what will happen? What does CPI have to do at that time?** To overcome this situation, **while using the Router palette function you must and should define the Default Route** — it catches everything that matches no condition, so the message always has a path to an end.

> **Practical tip:** don't leave the default route empty. Common choices: route it to an **Error End Event** (so unexpected data shows as Failed), an **Escalation End** (shows as Escalated — Day 16), or a branch that logs/sends an alert.

4. What If Multiple Conditions Are Satisfied?

Question ①: what is a Router? → to send the data in **one branch** based on a condition.

Question ②: what if multiple conditions are satisfied? Say you have **3 routing conditions and 1 default route**, and **all 3 conditions are satisfied** — where will the data go?

Scenario: Body → `<status>Active</status>` and header `RouterId > 1` (e.g., value 4) — **two conditions are satisfied.** What happens?

> **Answer: how many conditions are satisfied doesn't matter.** The **message never goes to multiple branches** — the Router routes the data to **only one branch**, **irrespective of how many conditions are matched or not matched.**

> **Which branch exactly? (verified):** SAP's guidance says three things about the Router — **a default route is always required**, **the order of the conditions is important**, and **only one branch gets processed.** Routes are checked in the order listed, and the **first one whose condition is true** is taken — so put your most specific condition first and the default last.

5. Router Limitation — More Than One Record

Payload with 2 records: Emp ID 4 with Status "Active", and Emp ID 5 with Status "Inactive". The Router **cannot handle** this situation if we send **more than 1 record** in one message.

> **Why:** the Router decides **once per message**, not once per record — so it can't send record 4 one way and record 5 another. **Added solution: split first** (General or Iterating Splitter — Day 9, and the next topic in your curriculum), so each record becomes its own message, **then** route each one.

Part B — Multicast

6. What Is Multicast?

> **Multicast:** if you want to send the data to **multiple receivers at the same time**, we use the Multicast palette function.

Example requirement: we have **5 receivers**, and we need to send the **same data** to all 5 receivers **without any condition** → we use **Multicast**.

Multicast = same data → more than one receiver → sent at the same time.

7. Two Types of Multicast

Type	How It Works	When to Use

Parallel Multicast	Copies go to all branches **at the same time**	Multiple receivers with **no dependency** between them — "you call multiple receivers at the same time"
Sequential Multicast	Copies go **one by one, in a fixed order**	**1st receiver → depends → 2nd receiver** — the second must only be called after the first

Real-time rule: parallel vs. sequential **depends on the requirement**. Multiple receivers, no dependency → Parallel. First receiver, *then* second receiver → Sequential.

How to Define the Order in Sequential Multicast

Routing Sequence → Move Up / Move Down. In the Multicast step, branches are named Branch 1, Branch 2, … in the order you drew them; for a Sequential Multicast you open **Routing Sequence** and use **Move Up / Move Down** to set the order. (For Parallel Multicast no ordering is needed.)

8. The Two Very Important Multicast Questions

Question ① — What happens when the 1st receiver fails?

Multicast Type	If Branch 1 (1st receiver) fails, does data go to the 2nd receiver?
Sequential	**No** — it will **not** go to the second receiver; processing stops at the failure
Parallel	**Yes** ✓ — the other branches still receive and process the message

(This is the key practical difference: sequential = fail-fast chain; parallel = independent branches.)

Question ② — Can a header/property created in Branch 1 be used in Branch 2?

Multicast Type	Header / property created in Branch 1 — usable in Branch 2?
Sequential	**No**
Parallel (PoC — proof of concept shown in class)	**No**

> **Why (the reason behind the answer):** Multicast sends **copies of the same message** to each branch. Every branch works on **its own copy of the original message**, so changes made in one branch (new headers, properties, a changed body) are **not visible** in the others — even in Sequential mode. If Branch 2 needs something, set it **before** the Multicast step so every copy carries it.

9. Related Patterns (Added)

- **Merging branches back:** if the branches must come back together after a Multicast, use a **Join** (and **Gather** to combine the messages) — Day 9's routing list.
- **Recipient list pattern (SAP's own reference design):** a **Multicast** sends a copy to every branch, and a small **Router/Filter** in each branch decides whether that branch should really process it — the default route leads to an End event that simply discards the message. This gives you "send to *some* of many receivers based on content".

10. Router vs. Multicast — Side by Side

	Router	Multicast
Purpose	Choose **one** path based on a condition	Send **copies** to **all** paths
Conditions needed?	Yes (XML or non-XML)	No
Branches processed	Only **one**	**All** (parallel or in sequence)
Default route	**Mandatory**	Not applicable
Order matters?	Yes — first true condition wins	Only for Sequential (Routing Sequence)
Typical use	Status active → A, else → B	Same order data to ERP, warehouse and email at once

> **One line I couldn't interpret:** under the Sequential Multicast "Routing Sequence → Move Up / Move Down" note you wrote an arrow to **"filter ✓"**. It may be a pointer to the Filter step (the next topic) or to the recipient-list pattern in section 9 — I'd rather

> flag it than guess. Ask your trainer what was meant.

11. Quick Interview Answers

Question	Short Answer
What is a Router?	Sends the message to **one** branch based on a condition
Why is the default route mandatory?	Every process must end; the default route catches data that matches no condition
If three conditions are true, where does the message go?	To **one** branch only — the first true condition in order
Can the Router use non-XML conditions?	Yes — headers/properties with `${...}`, operators like =, !=, contains, in, regex, and, or
Router with a 2-record message?	It routes the whole message once — split first, then route
Parallel vs. Sequential Multicast?	Parallel: all branches at once, independent. Sequential: one by one in a set order
Branch 1 fails — what happens to Branch 2?	Sequential: not executed. Parallel: still executed
Can Branch 2 use a header created in Branch 1?	No — each branch has its own copy of the message